Environment Characteristics
(Nature of Environment in which Agent would work)

- In the context of AI and rational agents, the environment plays a crucial role in determining how an agent should be designed and how it operates.
- Here are the key characteristics of environments:
  - Fully Observable vs. Partially Observable
  - Deterministic vs. Stochastic
  - Episodic vs. Sequential
  - Static vs. Dynamic
  - Discrete vs. Continuous
  - Single-agent vs. Multi-agent

Fully Observable vs. Partially Observable

- Fully Observable: The agent can access the complete state of the environment at each point in time.
- Example: A chess game where the entire board is visible to the player.

- Partially Observable: The agent cannot access all aspects of the current state.
- Example: A poker game where a player can't see opponents' cards.

Deterministic vs. Stochastic

- Deterministic: The next state of the environment is completely determined by the current state and the agent's action.
- Example: A simple puzzle game where each move has a predictable outcome.

- Stochastic: There's an element of randomness or uncertainty in how the environment evolves.
- Example: Weather forecasting, where multiple factors can influence outcomes unpredictably.

Episodic vs. Sequential

- Episodic: The agent's experience is divided into atomic episodes with no link between the performances in different scenarios.
- Example: An image classification system where each image is classified independently.

- Sequential: Current decisions affect future situations.
- Example: A chess game where early moves influence later game states.

Static vs. Dynamic

- Static: The environment doesn't change while the agent is deliberating.
- Example: A crossword puzzle, where the puzzle doesn't change as you think.

- Dynamic: The environment can change while the agent is thinking.
- Example: Real-time strategy games where opponents make moves continuously.

Another example: off-line route planning vs. on-board navigation system

Discrete vs. Continuous

- Discrete: There are a finite number of distinct states and actions.
- Example: A tic-tac-toe game with a limited number of board positions.

- Continuous: State and action variables take on values in a continuous range.
- Example: A self-driving car where speed and direction can vary continuously.

Single-agent vs. Multi-agent

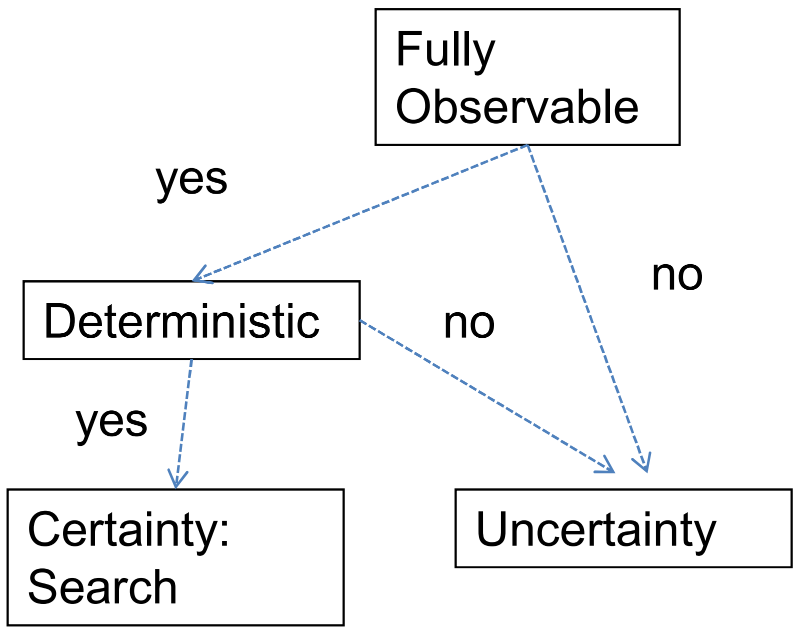
- Single-agent: Only one agent operates in the environment.
- Example: A robot vacuum cleaner working alone in a house.

- Multi-agent: Multiple agents interact in the environment.
- Example: A soccer-playing robot in a team of robots competing against another team.

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Interactive English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Figure 2.6 Examples of task environments and their characteristics.

Choice under (Un)certainty



Homework

For each of the following Agent’s, Describe the environment characteristics (6 properties):

- Playing soccer.
- Exploring the subsurface oceans of Titan.
- Shopping for used AI books on the Internet.
- Playing a tennis match.
- Practicing tennis against a wall.
- Performing a high jump.
- Knitting a sweater.
- Bidding on an item at an auction.